

Society for Computer Technology & Research's

PUNE INSTITUTE OF COMPUTER TECHNOLOGY, Pune.

DEPARTMENT OF INFORMATION TECHNOLOGY

DATA STRUCTURES MINI-PROJECT GROUP REPORT

Class: SY-09

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Mini-project Title: Memory Block Allocation and Deallocation Simulator

An operating system allocates memory blocks to processes and reclaims them when processes finish.

Create a memory-management simulator in which each linked-list node stores a block's start address, size, and allocation status. Implement first-fit allocation, deallocation, splitting of free blocks, merging adjacent free blocks, and display of the memory map after every operation. Report insufficient-memory conditions and demonstrate external fragmentation

Problem statement:

Create a memory-management simulator in which each linked-list node stores a block's start address, size, and allocation status. Implement first-fit allocation, deallocation, splitting of free blocks, merging adjacent free blocks, and display of the memory map after every operation. Report insufficient-memory conditions and demonstrate external fragmentation

Objectives:

1. Efficient Memory Allocation: Allocate memory to processes using the First Fit strategy.

2. Memory Deallocation: Release memory occupied by completed processes.

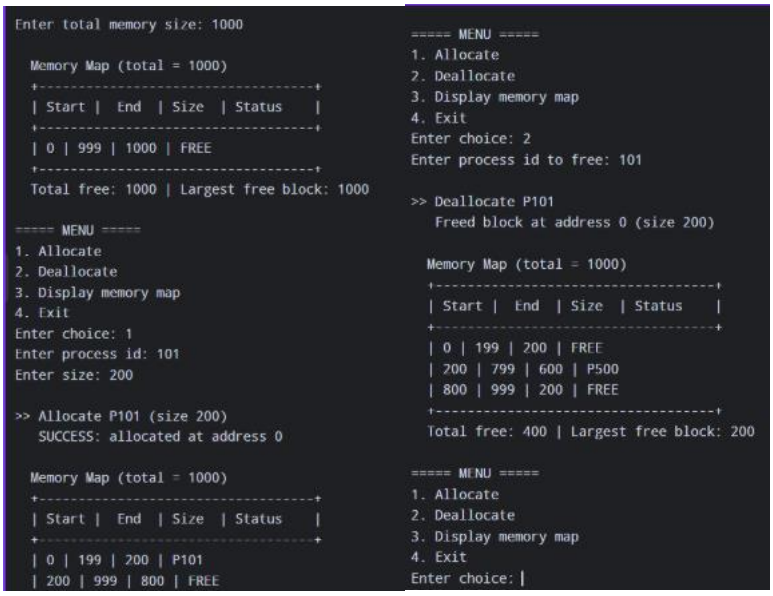
3. Memory Merging: Combine adjacent free blocks to reduce fragmentation.

Data structures used:

Doubly linked list Forward and backward traversal: We can move through the data in both directions using next and previous.

Implementation (Programming language / tools): C++

Implementation output:



Conclusion:

- Understood the working and implementation of a Doubly Linked List.
- Learned how to perform insertion, deletion, searching, and traversal operations.
- Understood the importance of `next` and `prev` pointers in connecting nodes.
- Learned how dynamic memory allocation helps manage data efficiently.